

Scalability & Future Plan

Date	15 March 2024 2 July 2026
Team ID	XXXXXX EduGenie - Google Gemini Powered Learning Assistant
Project Name	XXXXXX EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

JSON flat-file storage is

Risk of data loss/corruption under multi-user load

High

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
	No user authentication or authorization Single-process Uvicorn server, no caching layer	Cannot track individual learning history or restrict access	High Medium
1	Duplicate router-based and direct API endpoints	Increases maintenance overhead and API surface confusion	Medium
2			
3			

Scalability Plan:

Single-process Uvicorn, JSON files

Deploy behind Gunicorn + multiple Uvicorn workers; add a reverse proxy / load balancer

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
	Data	JSON flat files (data/*.json)	Migrate to PostgreSQL or MongoDB while preserving the existing db_manager.py interface
1	User Load Performance	No caching; every request hits Gemini/T5 Gemini/T5	Add response caching for repeated topics; batch quiz generation requests
2	Data Storage Security	No auth, open CORS	Add JWT-based authentication, restrict CORS origins, encrypt sensitive data at rest
3	Performance		
4	Security		

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2			
Phase 3			
Phase 4			

JSON flat-file storage is not safe under concurrent writes	Risk of data loss/corruption under multi-user load	High
No user authentication or authorization	Cannot track individual learning history or restrict access	High
Single-process Uvicorn server, no caching layer	Limited throughput under heavy concurrent traffic	Medium
Duplicate router-based and direct API endpoints	Increases maintenance overhead and API surface confusion	Medium

User Load	Single-process Uvicorn, JSON files	Deploy behind Gunicorn + multiple Uvicorn workers; add a reverse proxy / load balancer
Data Storage	JSON flat files (data/*.json)	Migrate to PostgreSQL or MongoDB while preserving the existing db_manager.py interface
Performance	No caching; every request hits Gemini/T5 Gemini/T5	Add response caching for repeated topics; batch quiz generation requests
Security	No auth, open CORS	Add JWT-based authentication, restrict CORS origins, encrypt sensitive data at rest